

SERVICE GUIDE FOR SPRING HANGERS & SUPPORTS



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Contents

- 1.0 Introduction
- 2.0 Importance
- 3.0 Design Process
- 4.0 Variable Spring Hangers
- 5.0 Constant Spring Hangers
- 6.0 Inspection of Spring Hangers
 - 6.1 Inspection after Installation and Prior to Hydrostatic Testing
 - 6.2 Inspection after Hydrostatic Testing and Prior to Heating Up
 - 6.3 Inspection of Service Spring Hangers
- 7.0 General Notes & Guidelines
- 8.0 Points to Remember

1.0 Introduction

This guide presents general information about pipe hanger and support systems, its types and how to inspect the pipe hangers.

According to the relevant codes and standards, Pipe hanger systems (primary and secondary) are divided into two groups: spring and rigid.

Based on whether the support is installed above or below the pipe, they are further classified as spring/rigid hangers (top-mounted) and spring/rigid supports (base-mounted).

2.0 Importance

Pipe supports are engineered to:

- Support the dead weight of the pipe, insulation, fluid and in-line components such as valves, flow meters and instrumentation.
- Restrain thermal pipe movement, as desired, from ambient to operating condition.
- Restrain pipe movement created by surge, steam hammer, water hammer or other flow-caused large dynamic loads.
- Restrain pipe movement caused by operational flow-induced vibrations such as vane passing pulsations, vortex shedding and other sources
- Restrain pipe movements caused by outside sources, such as wind and seismic events
- Provide stability to the pipe, particularly on long risers and long horizontal runs.
- Minimize sag to acceptable levels between supports, which may be limited to assure full drainage at all times.



A pipe support failure can cause catastrophic failure of a pipe, collateral damage to the plant, fires, explosions, and injury or death to personnel.

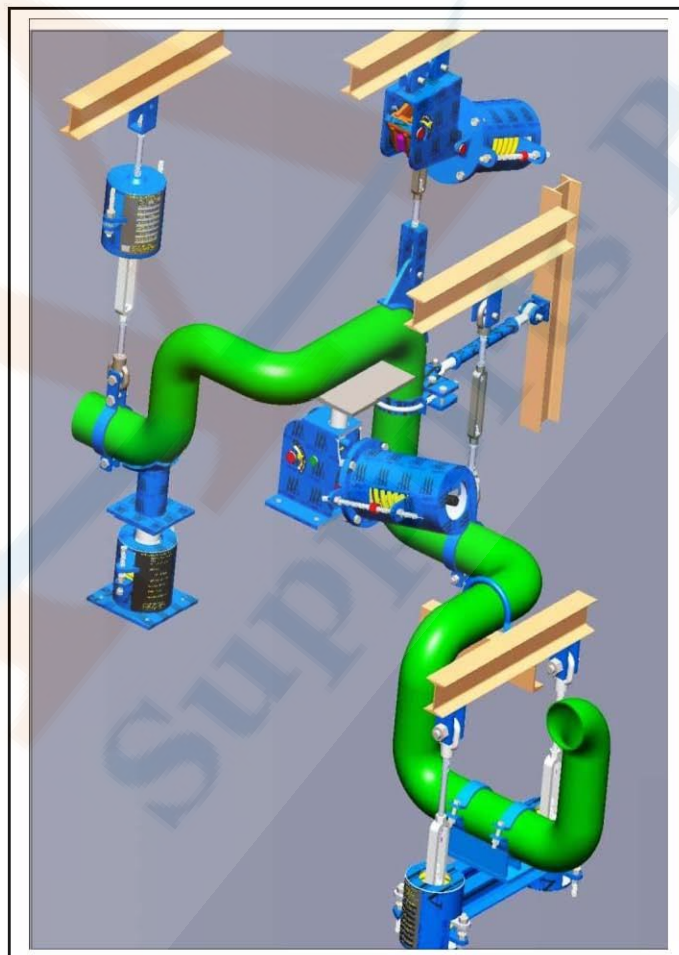
Pipe support damage can affect plant availability and may create serious conditions that could result in injury or death.

The main goal of spring support inspections is to help engineers and designers recognize potential problems and address them before they become catastrophic issues.

3.0 Design Process

Before routing a pipe, there should be a clear understanding of the process conditions; the expected loading conditions; acceptable areas for pipe per process safety and fire constraints; client expectations for access for maintenance, inspection and operations; an approximate layout of equipment; and a concept of the structural supports that will be available.

The initial plant layout approximately locates equipment, horizontal and vertical pipe chases, cable tray chases, walkways, crane access and other major considerations. General steel and concrete arrangement is based upon these layouts.



Approved pipe support spacing tables can be used for initial routing. The acceptable distance between pipe and steel is affected by the size and type of supports, along with the expected horizontal movements.

Proper planning and early decision-making in the design process will make the final support design drawings a relatively straightforward process. However, a poorly conceived and reviewed pipe support layout may result in significant difficulties in finalizing the steel design

and even in finding space to install the pipe supports.

While some of the failures described are the result of faulty fabrication, incorrect installation and unusual operating conditions – many failures are a direct result of faulty design practice.

4.0 Variable Spring Hangers

The design of a variable spring is based on:

	➤ Optimum load – usually set at operating or hot load ➤ Travel (mm/ inch) – the calculated vertical travel movement from ambient to operating condition, or cold to hot load movement (Convention is movement down cold to hot is negative, movement up is positive) ➤ Spring Rate – kg/mm or N/mm ➤ Ambient Load – Cold Load ➤ Load Variation Percent – VF %
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Problems with variable springs can be similar to those associated with rigid rods, such as bowed rods, a swing angle greater than 4°, or corroded components. Other common issues include:

- Pipe and spring do not move as expected.
- If a spring is installed below the pipe (F-type or base-mounted support), the 4° limit is not applicable. For base-type supports, manufacturers typically recommend a maximum horizontal movement of 0.25 inch to maintain the intended load condition.
- Spring is bottomed out – once the spring is fully compressed, it acts as a rigid support.
- Spring is topped out – the spring is fully extended against the top of the can.

Since the spring is pre-loaded, the load in the spring decreases from the pre-load until it eventually reaches 0.0 kg.

It is possible for a spring to appear topped out while still supporting some load.



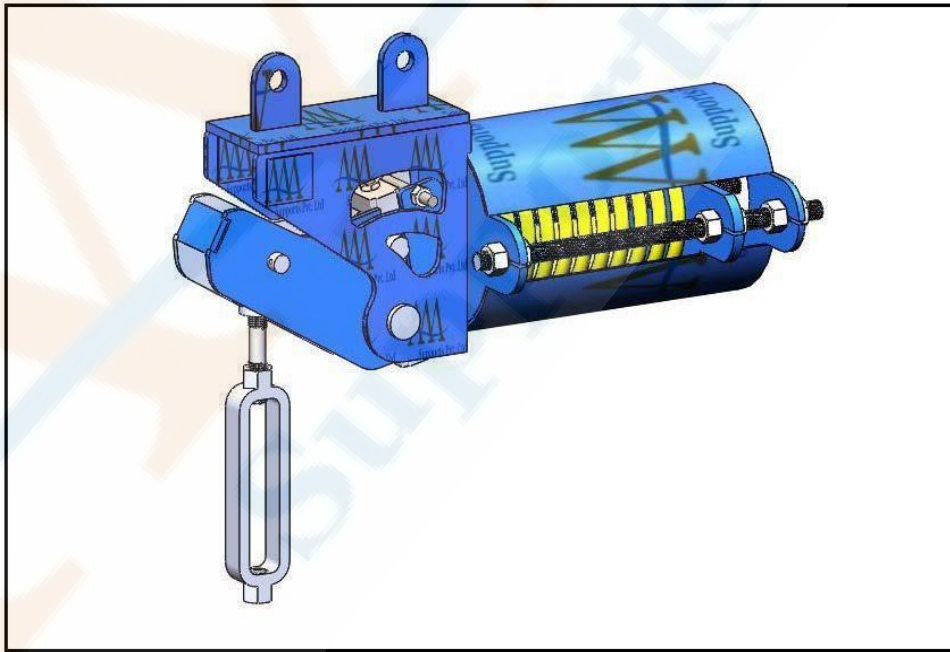
The following examples illustrate how improper installation and maintenance of variable spring supports can result in failure or damage to the spring supports or connected components.



5.0 Constant Spring Hangers

For the cases where the vertical movement is too great to use a variable spring hanger, pipe support vendors have developed constant spring supports. The name implies that the load is the same over the entire movement range of the hanger. However, according to MSS SP-58, the load may vary by $\pm 6\%$ over the full range and still be within specification. After the cans have aged, even the larger variations (more than 12%) have been documented.

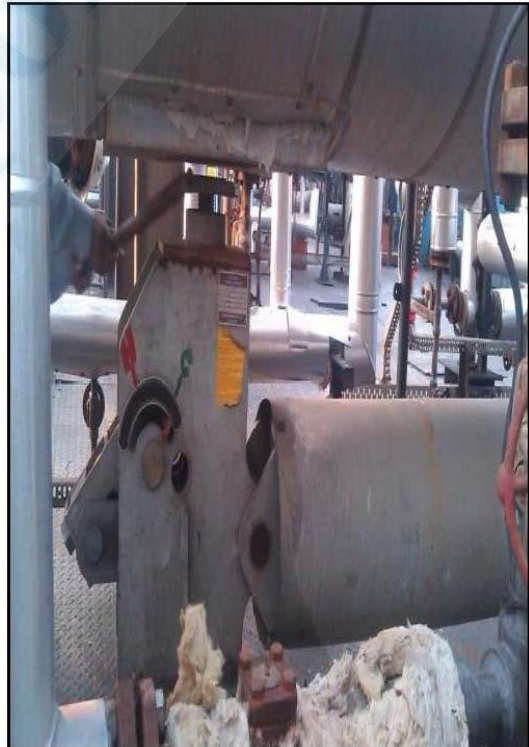
Most common supports have a helical coil (perhaps multiple helical coils), that act like a typical variable spring hanger. Traditionally, to reduce load variation through the range, lever arms and pivot pins transfer the load from the rod to the steel.



The physical structure may change from vendor to vendor but the operating principle is the same.

- As the pipe moves down, it pulls the load pivot pin downward.
- The main pivot rotates but does not move.
- The internal arms from the pivot pin to the end of the spring compress the spring.
- The load is adjusted by changing the distance from the main pivot to the movable spring pivot with the load adjustment bolt.

The following examples illustrate how improper installation and maintenance of constant spring supports can result in failure or damage to the spring supports or connected components.



6.0 Inspection of Spring Hangers

6.1 Inspection after Installation and Prior to Hydrostatic Testing

Each hanger assembly should be verified against the hanger drawing to ensure that all components have been installed and are in their proper position.

Temporary locking devices (travel stops) and other locking means furnished with spring units should be properly engaged.

6.2 Inspection after Hydrostatic Testing and Prior to Heating Up

Spring locking devices and any temporary hangers and supports required for hydrostatic testing shall be removed.

Spring hangers should be checked to ensure that their load/travel indicators are approximately in their cold position setting and that the pipe is at the proper elevation.

6.3 Inspection of Service Spring Hangers

Piping systems should be inspected in their entirety to verify that no interference exists between the pipe and/or pipe insulation and the building structure or equipment.

Spring hanger loads should be checked under both cold and hot conditions, and the readings obtained should be checked against the original cold and hot readings.

The plant should establish a program to have a periodic 'Inspection of critical piping support system and load travel indicator readings recorded and compared to the baseline set of readings. Spring hangers should be checked to ensure that their load/travel indicators are approximately in their hot position setting and that the pipe is at the proper elevation.

Line temperature and actual hot positions of the load/travel indicators on springs should be noted and recorded in a permanent formal log.

7.0 General Notes & Guidelines

- Any readjustment of the spring element shall be carried out only when the line is full of fluid, or an equivalent fluid of sufficient density, to balance the weight of the piping and the preset spring load.
- Adjustment of the hanger-type spring element is carried out by rotating the turnbuckle or adjustment nuts provided in the hanger rod.
- During hydraulic testing, flushing, or chemical cleaning of the pipeline, the spring must be

kept in the locked condition or protected against overloading due to the weight of the testing/flushing fluid by providing temporary supports.

- After readjustment, it is important to check whether sufficient range is available on scale for required movement of the pipe during operation.

8.0 Points to Remember

- Spring design will be based on the installation condition (operating load).
- Compactness of the unit – installation height should be kept to a minimum.
- In occasional cases, the pipe may move more than the operating movement. If the maximum deflection range is selected, the spring may have no further movement available and could fail. To address this, provide a “Cushion Range”; therefore, always select the “MID-RANGE” where appropriate.
- If the spring is designed for occasional loads (e.g., seismic, wind, etc.) and movement at the initial design stage, it may result in an over-design.
- For hanging spring supports, lateral movement (rod swing) should not exceed 4°.
- For bottom-type supports where horizontal movement of more than ½ inch is anticipated, Teflon-covered load pads should be specified.
- Always mention the hydro test load when ordering a spring. This will help the spring vendor in designing the spring locking arrangement.
- Standard inventory finish: Red Oxide Primer.
- Coils are supplied with a protective coating that protects against a wide range of corrosive environments and does not affect the flex life of the spring.
- Supports are fitted with nameplates marked with the installation and operating load, support reference mark, type and unique serial number.

-Thank You -